ENGINEERING CHANGE NOTE (ECN)

PROJECT: PROJECT-ADDMOTOR (revision of PROJECT-MYMILLI)
ORIGINATION: Prof. Kidd, Instructor (ECO 0.3.1, 3/1/2030, revised 4/1/2030)
PURPOSE OF CHANGE: Add a DC motor-driven fan with four speeds (OFF, LOW, MED, HIGH), driven by a PWM signal and controlled by one push button
DOCUMENT NUMBER: ECN0.3.1
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EXECUTIVE SUMMARY: This ECN implements ECO 0.3.1 by adding a DC motor-driven fan to the device. The fan has four states (OFF, LOW, MED anHIGH), and one push button steps it through OFF, LOW, MED, HIGH, MED, LOW, OFF and then repeats. An N-channel MOSFET switches the motor. Arduino pin 9 drives the MOSFET with a PWM signal made by Timer/Counter1 in Fast PWM mode (mode 14, TOP = 249, clock divided by 8), which gives 8 kHz. The duty cycles are 0% (OFF), 40% (LOW), 64% (MED) and 100% (HIGH). The button is read on pin 10 by polling, using the internal pull-up and a 30 ms software debounce that uses `mymilli()`. The fan code is in two new files, `fan.h` and `fan.cpp`, and the push button code is in two new files, `button.h` and `button.cpp`. Every register address and bit position they use is defined by a constant in those files. The sketch gains one task (`checkButton`) in the task table and the start-up calls for the button and fan modules at the end of `setup()`. The hardware makes the PWM, so the fan adds almost no processor load and is designed not to disturb the LEDs or the serial dialog. Timer/Counter0 and `mymilli()` are unchanged. The System Design Document is revised to Rev 4. Two educational constraints (EC.8, EC.9) and four tests (T.11 to T.14) are added, and the #include check of T.6 and T.7, T.8 and T.10 are changed.

HIGH-LEVEL DESCRIPTION: This ECN notifies stakeholders that PROJECT-ADDMOTOR adds a PWM-driven, push-button-controlled DC motor fan to PROJECT-MYMILLI. The two LED blink tasks, the serial dialog and the `mymilli()` time base of PROJECT-MYMILLI are unchanged.

REASON FOR CHANGE: ECO 0.3.1 states that the customer has added the requirement of a cooling fan on the device.

CUSTOMER REQUIREMENTS:

The customer requirements C1 through C6 of PROJECT-ASYNC are unchanged; see [1] REQUIREMENTS section, PART I, Table 1. ECO 0.3.1 adds seven customer requirements, numbered C6 through C12 in Table 1 of [5]: a motor-driven fan (C6), fan RPM controlled by a push button (C7), four speeds OFF, LOW, MEDIUM and HIGH (C8), each button hit cycling OFF, LOW, MEDIUM, HIGH, MEDIUM, LOW, OFF (C9), and the sound and air flow of LOW, MEDIUM and HIGH (C10 to C12). They are for reference only; the high-level technical requirements below carry the engineering work.

HIGH-LEVEL TECHNICAL REQUIREMENTS:

The high-level technical requirements HL.1 through HL.6 are identical to those of PROJECT-ASYNC; see [1] REQUIREMENTS section, PART II, Table 3. ECO 0.3.1 adds HL.7 through HL.18 (Table 2 of [5]). The educational constraints EC.1 through EC.7.2 are unchanged; see [1], [2] and [3]. Two educational constraints, EC.8 and EC.9, are added by this ECN. The ECO numbers them EC 1.2 and EC 1.3 (E.4 and E.5 in its Table 3); they are numbered EC.8 and EC.9 here because EC.4 and EC.5 already exist.

Changed Documents: provost.cmpe311.fall26.project#4 (System Design Document, Rev 4, revising provost.cmpe311.fall26.project#3 [3]). Deleted text is ~~struck out~~ and additions are **marked**, each with the date.

- 10/31/26 (REQUIREMENTS, Part II, Table 3) HL.7 to HL.13 added: DC motor-driven fan with an operating voltage between 3 V and 12 V (HL.7); fan speed between 1500 and 6500 RPM (HL.8); fan draws less than 4 A (HL.9); fan driven by a PWM signal of at most 5 VDC (HL.10); fan smaller than 4 in x 4 in x 3 in, not including cage (HL.11); RPM controlled by a single-pole, momentary-on, normally-off push button (HL.12); button and motor have no noticeable effect on the LEDs and serial monitor (HL.13). (ECO 0.3.1, Table 2)
- 10/31/26 (REQUIREMENTS, Part II, Table 3) HL.14 and HL.15 added: four speeds OFF, LOW, MEDIUM and HIGH (HL.14); each button hit cycles OFF, LOW, MEDIUM, HIGH, MEDIUM, LOW, OFF (HL.15). (ECO 0.3.1, Table 2)
- 10/31/26 (REQUIREMENTS, Part II, Table 3) HL.16 to HL.18 added with the corrected air flow ranges: LOW under 35 dB and ~~30-45~~ 0.30 to 0.45 CFM ± 0.05 CFM (HL.16); MEDIUM under 40 dB and ~~45-70~~ 0.45 to 0.70 CFM ± 0.05 CFM (HL.17); HIGH under 50 dB and ~~70-100~~ 0.70 to 1.00 CFM ± 0.05 CFM (HL.18). The struck-out ranges were deleted 4/1/2030 (twk) in [5].
- 10/31/26 (REQUIREMENTS, Part II, Table 4) EC.8 added: "The Evaluator must be able to successfully have The System perform the dialog in Table 4 and Table 5." (ECO 0.3.1, EC 1.2 and E.4, added 3/20/2030 by kidd; the earlier wording that named only Table 5 was deleted 3/20/2030 by twk). EC.9 added: "H.16, H.17, H.18 do not have to be explicitly measured. Just provide some evidence that these requirements have been satisfied through justified inference such as through calculations." (ECO 0.3.1, EC 1.3 and E.5)
- 10/31/26 (REQUIREMENTS, Table 4) Table 4 added: it maps each item of ECO 0.3.1 to where the revised document meets it.
- 10/31/26 (TESTING AND VALIDATION REQUIREMENTS) T.11 added: the motor has a rated voltage inside 3 V to 12 V, the supply limits current below 4 A, the fan fits in 4 in x 4 in x 3 in, the fan is driven only by the Timer/Counter1 PWM output on pin 9 at a 5 V logic level, and one single-pole, momentary-on, normally-off push button is connected between pin 10 and ground. Verifies HL.7, HL.9, HL.10, HL.11, HL.12. T.12 added: the fan steps OFF, LOW, MED, HIGH, MED, LOW, OFF and repeats, one step per button release. Verifies HL.14, HL.15, EC.8 (Table 5 of [5]). T.13 added: the LED dialog and the button presses run together with no noticeable change in LED timing or serial dialog. Verifies HL.13, EC.8 (Table 4 of [5]). T.14 added: motor speed, air flow and sound level checked by calculation. Verifies HL.8, HL.16, HL.17, HL.18, EC.9.
- 10/31/26 (TESTING AND VALIDATION REQUIREMENTS, T.8 and T.10) T.8 now also covers `fan.h`, `fan.cpp`, `button.h` and `button.cpp`: no call to `millis()`, `micros()` or `delay()`, and the button debounce uses `mymilli()` as its only time source. T.10 now also covers `fan.cpp` and `button.cpp`: every timer and port register address and bit position is defined by a constant, and no AVR-defined register name is present. T.1 through T.10 are re-run on the PROJECT-ADDMOTOR build.
- 10/31/26 (TEST PROCEDURES, T.6 and T.7) The search for #include found exactly one in the sketch, `#include "mymilli.h"`. It now finds only headers written by The Developer (`mymilli.h`, `fan.h` and `button.h`). Pass criterion: the only #include lines in the .ino file are those headers, and no library or scheduler header is included. Procedures for T.11 to T.14 added. The task count in T.7 changes from three to four, because the task table now holds `checkButton`.

- 10/31/26 (REQUIREMENTS TRACEABILITY) Rows added: HL.7, HL.9, HL.10, HL.11 and HL.12 verified by T.11; HL.8 by T.14; HL.13 by T.13; HL.14 and HL.15 by T.12; HL.16 to HL.18 by T.14; EC.8 by T.12 and T.13; EC.9 by T.14.
- 10/31/26 (SYSTEM OVERVIEW) Added: a task, `checkButton`, is added to the task table and passes each button release to the fan module, which sets the fan speed. The task table now holds four tasks. The dispatcher is unchanged because it counts the table entries itself.
- 10/31/26 (SOFTWARE DESIGN) Added: the fan module in two new files, `fan.h` and `fan.cpp`, and the button module in two new files, `button.h` and `button.cpp`; the `#include` lines for their headers at the top of the sketch and their start-up calls after `mymilliInit()` in `setup()`; timer selection (Timer/Counter1, which is free and has an output on pin 9); timer configuration (Fast PWM mode 14, TOP = ICR1 = 249, clock / 8, 8 kHz); the register constants and the hardware items they stand for; the duty cycle for each fan state; the button read and the 30 ms debounce; and the state machine, which changes state on a button release and remembers its direction so the two MED visits and the two LOW visits in the cycle are told apart.
- 10/31/26 (BILL OF MATERIALS) Added: DC motor with fan blade; IRF520N MOSFET; 330 Ω and 10 kΩ resistors; Schottky diode; push button; 5 V power module with a 9 V battery (Table 5).
- 10/31/26 (HARDWARE DESIGN) Added: DC motor with fan blade; N-channel MOSFET low-side switch (IRF520N); 330 Ω gate resistor; 10 kΩ gate pull-down resistor; Schottky flyback diode across the motor; push button; 5 V breadboard power module with a 9 V battery for the motor; pin 9 (fan PWM) and pin 10 (button) assigned. The Arduino, power module and MOSFET source share one ground. LED pins 2 and 3 and the serial pins are unchanged.
- 10/31/26 (DESIGN RATIONALE AND ASSUMPTIONS) Added: why Timer/Counter1 and pin 9; why Fast PWM at 8 kHz; why HIGH is OCR1A = TOP; why the fan changes state on a button release; why the button is polled; the 30 ms debounce; why the fan does not disturb the LEDs and serial dialog; the MOSFET choice; resources given up (Timer/Counter1, `analogWrite()` on pins 9 and 10, and the Servo library); and the assumptions behind the EC.9 calculations.
- 10/31/26 (TESTING AND VALIDATION, CALCULATIONS FOR EC.9) Added: the calculations of speed, air flow and sound level that justify HL.8 and HL.16 to HL.18 (Table 16).
- 10/31/26 (APPENDIX A, TEST EVIDENCE) Rows added for T.11 to T.14; evidence links point to the CMPE311Project4 repository.
- 10/31/26 (APPENDIX B, DESIGN CODE) PROJECT-MYMILLI source replaced by the PROJECT-ADDMOTOR source in seven files: the sketch, `mymilli.h`, `mymilli.cpp`, `fan.h`, `fan.cpp`, `button.h` and `button.cpp`. `mymilli.h` and `mymilli.cpp` are unchanged. The sketch adds the new `#include` lines, the `checkButton` task and the start-up calls.
- 10/31/26 (TITLE AND DATE, REFERENCES AND GLOSSARY, TESTING RESULTS) Document reference changed to provost.cmpe311.fall26.project#4; revision history, references and new terms added; results and evidence to be re-recorded for the PROJECT-ADDMOTOR build, with rows added for T.11 to T.14.

Sections not listed above (customer requirements C1 through C6, HL.1 through HL.6, EC.1 through EC.7.2, the two LED tasks, the serial dialog, the `mymilli()` time base, the LED wiring and pins, and T.1 through T.5) are unchanged and are cited to [1], [2] and [3] in the revised document.

Added Documents: ECN0.3.1, provost.cmpe311.fall26.project#4.ECN (this note), added 10/31/26. provost.cmpe311.fall26.project#4 (System Design Document, Rev 4), added 10/31/26. `fan.h`,

`fan.cpp`, `button.h` and `button.cpp`, and the updated PROJECT-ADDMOTOR sketch, added to the GitHub repository https://github.com/raheemprovost/CMPE311Project4 on 10/31/26.

Removed Documents: None.

Replaced Documents: None. provost.cmpe311.fall26.project#3 [3] remains the PROJECT-MYMILLI baseline, provost.cmpe311.fall26.project#2 [2] remains the PROJECT-CE baseline, and provost.cmpe311.fall26.project#1 [1] remains the PROJECT-ASYNC baseline.

Any Additional Design Considerations: Hardware change: a DC motor fan, a MOSFET driver, a push button and a 5 V motor supply are added to the board; the LEDs, 330 Ω LED resistors and pins 2 and 3 are as in [1]. Pin 9 and pin 10 are now reserved for the fan PWM output and the button. Timer/Counter1 is now reserved for the fan PWM, so `analogWrite()` on pins 9 and 10 and the Servo library cannot be used. Timer/Counter0 is unchanged and still belongs to `mymilli()`. Teams reusing the fan code must add `fan.h`, `fan.cpp`, `button.h` and `button.cpp` to the sketch folder, include their headers, call their start-up functions once in `setup()` after `mymilliInit()` (the button debounce reads `mymilli()`), and add a task that calls the button and fan update functions to the task table. The PWM frequency is 8 kHz (16 MHz ÷ 8 ÷ 250) and assumes a 16 MHz clock. The motor supply ground, the MOSFET source and the Arduino ground must be joined. The fan changes speed when the button is released, not when it is pressed. The IRF520N is not a logic-level MOSFET; it turns on far enough at 5 V gate drive for this small motor, and a logic-level part such as the IRLZ44N would turn on fully and can be fitted in its place. Air flow and sound level of LOW, MEDIUM and HIGH are justified by calculation, as EC.9 allows, and are not measured.

REFERENCES:

[1] provost.cmpe311.fall26.project#1, CMPE311 Project 1 design document (PROJECT-ASYNC), Revision 1, September 22, 2026.

[2] provost.cmpe311.fall26.project#2, CMPE311 Project 2 design document (PROJECT-CE), Revision 2, October 3, 2026.

[3] provost.cmpe311.fall26.project#3, CMPE311 Project 3 design document (PROJECT-MYMILLI), Revision 3, October 15, 2026.

[4] provost.cmpe311.fall26.project#4, CMPE311 Project 4 design document (PROJECT-ADDMOTOR), Revision 4, October 31, 2026.

[5] PROJECT: PROJECT-ADDMOTOR assignment, including Engineering Change Order ECO 0.3.1 (3/1/2030, revised 4/1/2030, requestor Prof. Kidd), CMPE311.